

# BODY PERFORMANCE ANALYTICS

## Intelligent Classification System

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Course Project Report  
Introduction to AI and ML

Under supervision of  
**Prof. /**

**DR .Sherif Ibrahim Morsy Hussein**

### Team Members

**Muhammad Sleem — Team Leader**

Essam Muhammad

Hazem Khalid

Muhammad Kamal

Muhammad Karem

*Submission: LMS Upload | PDF Report + Presentation + Source Code*

# 1. PROJECT OVERVIEW

## Body Performance Analytics and Intelligent Classification System

This report documents the complete machine learning project conducted by the team for the Introduction to AI and ML course. The project follows the three-part structure defined in the Course Project Handbook, covering data preparation, model training, and performance evaluation using the Body Performance Dataset.

The dataset contains physical fitness measurements from individuals undergoing performance evaluation, including 12 attributes such as age, gender, height, weight, body fat percentage, blood pressure readings, grip force, flexibility, sit-up count, and broad jump distance. The target variable 'class' categorises each participant's physical performance level.

### 1.1 Project Objectives

- Understand the structure and characteristics of the Body Performance Dataset
- Perform comprehensive Exploratory Data Analysis (EDA) including univariate, bivariate, and correlation analysis
- Identify and handle data quality issues: missing values, duplicates, and outliers
- Implement and compare five machine learning models: KNN, Decision Tree, SVM, XGBoost, and Neural Network
- Evaluate classification performance using accuracy, precision, recall, F1-score, and confusion matrix
- Build regression models predicting broad\_jump\_cm and evaluate with MSE, RMSE, and R<sup>2</sup>
- Compare model performance across different train-test splits (80:20, 70:30, 50:50) and K-Fold cross-validation

### 1.2 Tools and Libraries

| Library / Tool         | Purpose                                                 |
|------------------------|---------------------------------------------------------|
| Python 3               | Primary programming language                            |
| pandas                 | Data loading, manipulation, and analysis                |
| numpy                  | Numerical computations and array operations             |
| matplotlib / seaborn   | Data visualizations and plotting                        |
| scikit-learn           | ML models, preprocessing, metrics, and cross-validation |
| XGBoost                | Gradient boosting classification and regression         |
| PyTorch                | Custom Neural Network implementation                    |
| Google Colab / Jupyter | Development environment                                 |

## 2. TEAM ROLES AND RESPONSIBILITIES

Task Distribution Across Team Members

To ensure smooth collaboration and clear accountability, the project tasks were distributed according to each member's area of focus. The following cards detail the responsibilities assigned to each team member.

### Part 1 — Data Preparation & Exploratory Data Analysis

| Eissam Mahmoud <i>Data Overview &amp; Column Analysis</i> |                                                                                             |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| ✓                                                         | Dataset overview — rows (13,393), columns (12), column names, and sample rows               |
| ✓                                                         | Column understanding — data types (numeric vs categorical) for all 12 attributes            |
| ✓                                                         | Expected constraints and validation rules per column (e.g., age > 0, weight > 30 kg)        |
| ✓                                                         | Missing values analysis — count and percentage per column; defined imputation strategy      |
| ✓                                                         | Strategy documentation: median imputation for numeric columns, mode/Unknown for categorical |

| Hazem Khalid <i>Data Quality &amp; Statistical Analysis</i> |                                                                                                    |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| ✓                                                           | Duplicate detection — scanned for exact duplicate rows and reported findings                       |
| ✓                                                           | Data validity checks — flagged impossible values (negative weights, unrealistic ages, abnormal BP) |
| ✓                                                           | Univariate statistics — computed mean, median, standard deviation, minimum, maximum per column     |
| ✓                                                           | Distribution analysis — histograms for all 10 numeric features with mean-line annotations          |
| ✓                                                           | Boxplot visualizations for outlier detection; IQR-based outlier counts per column                  |
| ✓                                                           | Skewness analysis to identify right- or left-skewed distributions                                  |

## Part 2 — Machine Learning Model Training

### Muhammad Kamal *Classification Models (KNN, Decision Tree, SVM)*

- ✓ Implemented K-Nearest Neighbours classifier with tuning over  $k \in \{3, 5, 7, 9, 11, 15\}$
- ✓ Implemented Decision Tree classifier with `max_depth` tuning  $\in \{3, 5, 7, 10, 15, 20, \text{None}\}$
- ✓ Implemented Support Vector Machine with three kernels: linear, RBF, and polynomial
- ✓ Applied Standard Scaler preprocessing for distance-based models (KNN, SVM)
- ✓ Documented classification results: accuracy, precision, recall, F1-score for each configuration
- ✓ Identified best hyperparameter values based on weighted F1-Score on 80:20 split

### Muhammad Karem *Regression & Neural Network Models*

- ✓ Implemented Linear Regression as baseline model for predicting `broad_jump_cm`
- ✓ Implemented Neural Network regressor (MLPRegressor) with architectures: (50,), (100,), (50,25), (100,50)
- ✓ Implemented custom Neural Network classifier (MLPClassifier) with architectures up to (128, 64, 32)
- ✓ Built a PyTorch-based flexible NeuralNet class supporting variable hidden layer configurations
- ✓ Applied early stopping and validation fraction to prevent overfitting
- ✓ Ensured full reproducibility by setting `random_state=42` and `np.random.seed(42)`

Part 3 — Performance Evaluation & Model Testing

| Muhammad Sleem <i>Team Leader — Evaluation &amp; Coordination</i> |                                                                                                |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| ✓                                                                 | Coordinated cross-validation experiments across splits: 80:20, 70:30, 50:50, and K-Fold (k=10) |
| ✓                                                                 | Evaluated classification metrics: accuracy, precision, recall, F1-score, confusion matrix      |
| ✓                                                                 | Evaluated regression metrics: MSE, RMSE, R <sup>2</sup> for all regression models              |
| ✓                                                                 | Compiled model comparison dashboard comparing all models on 80:20 split                        |
| ✓                                                                 | Identified best-performing model for classification (XGBoost) and regression                   |
| ✓                                                                 | Summarized key findings in the conclusion section of the notebook                              |
| ✓                                                                 | Led final demonstration and compiled the analytical report                                     |

Shared Responsibilities

| Task                                                     | Responsible Members             |
|----------------------------------------------------------|---------------------------------|
| Final analytical report — each writes their section      | All Team Members                |
| Heatmaps, scatter plots, confusion matrix visualizations | Hazem Khalid & Eissam Mahmoud   |
| Presentation slides (model results and comparisons)      | Muhammad Kamal & Muhammad Karem |
| Report compilation, presentation polish, demo lead       | Muhammad Sleem (Team Leader)    |

## 3. PART 1 — DATA PREPARATION & EDA

Eissam Mahmoud & Hazem Khalid

### 3.1 Dataset Overview (Essam Muhammad)

The Body Performance Dataset was loaded from bodyPerformance.csv. The following basic facts were established:

- Total rows: 13,393 participant records
- Total columns: 12 attributes (10 numeric, 2 categorical)
- Column names: age, gender, height\_cm, weight\_kg, body\_fat\_%, diastolic, systolic, gripForce, sit\_and\_bend\_forward\_cm, sit-ups counts, broad\_jump\_cm, class

### 3.2 Column Understanding (Eissam Mahmoud)

| Column                  | Type        | Description                | Constraints   |
|-------------------------|-------------|----------------------------|---------------|
| age                     | Numeric     | Age of participant         | 10 – 100      |
| gender                  | Categorical | Participant gender (M/F)   | M or F        |
| height_cm               | Numeric     | Height in centimeters      | 140 – 220     |
| weight_kg               | Numeric     | Body weight in kilograms   | 30 – 200      |
| body_fat_%              | Numeric     | Body fat percentage        | 5 – 50        |
| diastolic               | Numeric     | Diastolic blood pressure   | 50 – 110      |
| systolic                | Numeric     | Systolic blood pressure    | 80 – 200      |
| gripForce               | Numeric     | Hand grip strength (kgf)   | 0 – 300       |
| sit_and_bend_forward_cm | Numeric     | Flexibility measure (cm)   | No hard limit |
| sit-ups counts          | Numeric     | Sit-up repetitions         | 0 – 200       |
| broad_jump_cm           | Numeric     | Explosive power (cm)       | 0 – 350       |
| class                   | Categorical | Performance category (A–D) | A, B, C, D    |

### 3.3 Data Type Verification (Essam Muhammad)

All 10 numeric columns were verified to have correct int64 or float64 dtypes. The gender and class columns were stored as object (string) type, which is appropriate for categorical variables. No type mismatches were detected that required correction.

### 3.4 Missing Values Analysis (Essam Muhammad)

A complete missing-value audit was performed on all 12 columns. The dataset showed zero missing values across all attributes, meaning no imputation was required. The strategy defined

for future reference was median imputation for numeric columns and mode/Unknown for categorical columns.

### 3.5 Duplicate Detection (Hazem Khalid)

The dataset was scanned for fully duplicated rows using pandas `.duplicated()`. A small number of duplicate records were found and reported. The decision was made to drop duplicates before model training to prevent data leakage and training bias, resulting in a clean dataset of 13,393 unique records.

### 3.6 Data Validity Checks (Hazem Khalid)

Each numeric column was checked against physiologically reasonable bounds:

| Column         | Lower Bound | Upper Bound | Status |
|----------------|-------------|-------------|--------|
| age            | 10          | 100         | OK     |
| weight_kg      | 30          | 200         | OK     |
| height_cm      | 140         | 220         | OK     |
| body_fat_%     | 5           | 50          | OK     |
| diastolic      | 50          | 110         | OK     |
| systolic       | 80          | 200         | OK     |
| gripForce      | 0           | 300         | OK     |
| sit-ups counts | 0           | 200         | OK     |
| broad_jump_cm  | 0           | 350         | OK     |

No records violated the defined bounds. All measurements were within physiologically plausible ranges.

### 3.7 Univariate Statistical Analysis (Hazem Khalid)

Descriptive statistics (mean, median, std, min, max) and skewness were computed for all numeric columns. The class distribution was roughly balanced across A, B, C, and D categories. Gender distribution showed more male participants than female.

### 3.8 Distribution Analysis (Hazem Khalid)

Histograms were generated for all 10 numeric features with mean lines overlaid. Key observations:

- Age showed a right-skewed distribution, with most participants between 20–50 years
- Height and weight distributions approximated normality
- gripForce and broad\_jump\_cm exhibited slight right skew, indicating some high-performance outliers

- `body_fat_%` showed a broader spread suggesting varied fitness levels

### 3.9 Outlier Detection (Hazem Khalid)

Boxplots were generated for each numeric feature. IQR-based outlier counts were recorded. The decision was made to cap or retain outliers (rather than remove them) where they likely represented genuine physiological variation, preserving data volume for model training.

### 3.10 Correlation Analysis (Hazem Khalid)

A correlation heatmap was generated for all numeric features. Key findings:

- `gripForce` showed strong positive correlation with sit-ups counts and `broad_jump_cm`
- `body_fat_%` showed negative correlation with performance-related features
- `age` showed moderate negative correlation with explosive power and grip strength
- Height and weight were positively correlated, as expected

### 3.11 Final EDA Summary

Five key insights discovered:

- Grip strength and sit-up count are the strongest predictors of performance class
- Body fat percentage has a clear inverse relationship with performance
- Age negatively affects most performance metrics
- Male participants tended to have higher grip force and broader jump distances
- The class distribution was reasonably balanced, reducing class imbalance concerns

Recommended preprocessing steps applied: duplicate removal, feature scaling (Standard-Scaler) for distance-based models, and gender label encoding (M=1, F=0).



## 4. PART 2 — MACHINE LEARNING MODEL TRAINING

Muhammad Kamal & Muhammad Kareem

### 4.1 Data Preprocessing

Before model training, the following preprocessing steps were applied:

- Duplicate rows were removed from the dataset
- Gender encoded: M  $\rightarrow$  1, F  $\rightarrow$  0
- Class encoded: A  $\rightarrow$  0, B  $\rightarrow$  1, C  $\rightarrow$  2, D  $\rightarrow$  3 using Label Encoder
- Feature matrix X constructed from all numeric columns plus gender\_encoded
- Classification target: class\_encoded; Regression target: broad\_jump\_cm
- Standard-Scaler applied to all features for distance-sensitive models (KNN, SVM, NN)
- Random seed fixed at 42 for full reproducibility

### 4.2 Classification Models (Muhammad Kamal)

#### 4.2.1 K-Nearest Neighbours (KNN)

Six values of k were tested: 3, 5, 7, 9, 11, and 15. Standard-Scaler was applied before fitting. The 80:20 train-test split was used for initial hyperparameter comparison.

| k Value | Accuracy | F1-Score |
|---------|----------|----------|
| k = 3   | 62.7%    | 0.627    |
| k = 5   | 62.7%    | 0.627 ★  |
| k = 7   | 62.5%    | 0.624    |
| k = 9   | 62.1%    | 0.620    |
| k = 11  | 61.8%    | 0.617    |
| k = 15  | 61.2%    | 0.611    |

Best k was selected based on highest weighted F1-Score on the 80:20 split.

#### 4.2.2 Decision Tree

Seven max\_depth values were tested: 3, 5, 7, 10, 15, 20, and None (fully grown). No feature scaling required for tree-based models.

| max_depth | Accuracy | F1-Score | Notes                   |
|-----------|----------|----------|-------------------------|
| 3         | 58.3%    | 0.578    | Underfits — too shallow |
| 5         | 63.1%    | 0.629    | Good balance            |
| 7         | 66.8%    | 0.666    | Near optimal            |

|      |       |         |                   |
|------|-------|---------|-------------------|
| 10   | 68.0% | 0.681 ★ | Best depth        |
| 15   | 67.5% | 0.673   | Starts to overfit |
| 20   | 66.2% | 0.660   | Overfitting       |
| None | 64.8% | 0.647   | Max overfit       |

### 4.2.3 Support Vector Machine (SVM)

Three kernel types were evaluated on the scaled dataset: linear, RBF, and polynomial.

| Kernel | Accuracy | F1-Score | Notes                        |
|--------|----------|----------|------------------------------|
| Linear | 67.2%    | 0.671    | Fastest training             |
| RBF    | 68.9%    | 0.690 ★  | Best classification boundary |
| Poly   | 67.8%    | 0.676    | Higher training time         |

## 4.3 XGBoost Classifier (Muhammad Sleem)

XGBoost was implemented as an additional classification model, extending beyond the base handbook requirements. Multiple hyperparameter configurations were tested (`n_estimators`, `learning_rate`, `max_depth`). XGBoost achieved the highest classification performance overall.

## 4.4 Neural Network Classifier (Muhammad Kmal)

MLPClassifier from scikit-learn was used with five hidden layer architectures tested:

- (64,) — single hidden layer, 64 neurons
- (128,) — single hidden layer, 128 neurons
- (64, 32) — two hidden layers
- (128, 64) — two hidden layers
- (128, 64, 32) — three hidden layers

ReLU activation, Adam optimizer, early stopping with 10% validation split, max 500 iterations. A custom PyTorch NeuralNet class was also implemented for flexible architecture exploration.

## 4.5 Regression Models (Muhammad Karem)

### 4.5.1 Linear Regression

Ordinary least squares regression was used as the interpretable baseline for predicting `broad_jump_cm`. Results were reported as MSE, RMSE, and  $R^2$  on the 80:20 test split.

### 4.5.2 Neural Network Regressor

MLPRegressor was tested with architectures (50,), (100,), (50, 25), and (100, 50). Early stopping was applied to prevent overfitting on the regression target.

### 4.5.3 XGBoost Regressor

XGBoost was also applied to the regression task. On the 80:20 split it achieved MSE = 365.47, RMSE = 19.12,  $R^2 = 0.770$ . Note: the Neural Network regressor outperformed XGBoost on this task ( $R^2 = 0.7816$ ).

## 4.6 Cross-Validation Experiments (Muhammad Sleem)

All models were evaluated under three train-test split ratios and 10-Fold cross-validation:

| Experiment    | Train Set | Test Set       | Purpose                  |
|---------------|-----------|----------------|--------------------------|
| 80:20 Split   | 80%       | 20%            | Primary evaluation split |
| 70:30 Split   | 70%       | 30%            | More test data           |
| 50:50 Split   | 50%       | 50%            | Equal split stress test  |
| K-Fold (k=10) | 90%       | 10% (rotating) | Most robust estimate     |

K-Fold cross-validation used scikit-learn pipelines (`make_pipeline`) to ensure `StandardScaler` was applied inside each fold, preventing data leakage.

## 5. PART 3 — PERFORMANCE EVALUATION

Muhammad Sleem (Team Leader)

### 5.1 Classification Metrics

All classification models were evaluated on the 80:20 holdout test set using:

| Metric               | Description                                                    |
|----------------------|----------------------------------------------------------------|
| Accuracy             | Proportion of correctly classified samples                     |
| Precision (weighted) | Positive predictive value, weighted by class support           |
| Recall (weighted)    | True positive rate, weighted by class support                  |
| F1-Score (weighted)  | Harmonic mean of precision and recall — primary ranking metric |
| Confusion Matrix     | Per-class breakdown of correct and incorrect predictions       |

### 5.2 Regression Metrics

| Metric               | Formula / Description                                           |
|----------------------|-----------------------------------------------------------------|
| MSE                  | Mean of squared prediction errors — penalizes large errors more |
| RMSE                 | Square root of MSE — same units as target variable (cm)         |
| R <sup>2</sup> Score | Proportion of variance explained; 1.0 = perfect prediction      |

### 5.3 Best Model Identification

After comparing all models, the best performers were identified as follows:

| Task           | Best Model               | Key Metric                                                         |
|----------------|--------------------------|--------------------------------------------------------------------|
| Classification | XGBoost Classifier       | F1-Score = 0.749, Accuracy = 75.1% on 80:20 split                  |
| Regression     | Neural Network Regressor | Lowest MSE (346.86), RMSE (18.62), highest R <sup>2</sup> = 0.7816 |

For classification, XGBoost achieved the highest accuracy (75.1%) and F1-Score (0.749) on the 80:20 split, outperforming KNN (62.7%), Decision Tree (68.0%), SVM (68.9%), and Neural Network (70.5%). For regression, the Neural Network achieved the best R<sup>2</sup> Score (0.7816), followed by Linear Regression (0.773) and XGBoost (0.770).

### 5.4 Required Visualizations

The following visualizations were produced in the notebook, with interpretation included:

- Histogram distributions for all 10 numeric features with mean-line annotations
- Boxplots per feature for outlier detection, styled with cosmic theme palette
- Correlation heatmap using seaborn with annotated values
- Scatter plots: Height vs Weight, Age vs Grip Force, Body Fat vs Class, Broad Jump vs Grip Force
- Categorical frequency plots: Gender distribution and Performance Class distribution
- Confusion matrix visualization for the best classification model
- Model comparison bar charts: F1-Score (classification) and  $R^2$  Score (regression)

## 6. GRADING RUBRIC

Total Points: 20

| Section                | Points    |
|------------------------|-----------|
| Data Preparation       | 4         |
| EDA Analysis           | 4         |
| Model Training         | 4         |
| Cross Validation       | 3         |
| Performance Evaluation | 3         |
| Report Quality         | 1         |
| Presentation           | 1         |
| <b>TOTAL</b>           | <b>20</b> |

## 7. ACADEMIC INTEGRITY DECLARATION

In accordance with the Academic Integrity requirements stated in the Course Project Handbook:

- This submission is the original work of our team.
- All team members contributed to their assigned sections as described in the Team Roles document.
- No content has been copied from other teams or external sources without attribution.
- AI tools were used to assist with code structuring, explanation, and report drafting. This is declared here as required.
- The submitted notebook executes without errors, and all results are reproducible using the provided source code with `random_state=42`.

### Team Signatures

| Name           | Role        | Contribution                                                |
|----------------|-------------|-------------------------------------------------------------|
| Muhammad Sleem | Team Leader | Part 3: Evaluation, Cross-Validation, Report Compilation    |
| Essam Mohamed  | Member      | Part 1: Dataset Overview, Column Analysis, Missing Values   |
| Hazem Khalid   | Member      | Part 1: Duplicates, Validity, Stats, Distribution, Outliers |
| Muhammad Kamal | Member      | Part 2: KNN, Decision Tree, SVM Implementation              |
| Muhammad Karem | Member      | Part 2: Linear Regression, Neural Networks, Reproducibility |